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The spread of *Fagus grandifolia* across eastern North America during the last 18 000 years

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ABSTRACT. The spread of *Fagus grandifolia* in eastern North America illustrates the assumptions behind the mapping of species distribution changes. Recognition of the distinction between population expansion and arrival of a species is crucial for interpreting pollen diagrams. *F. grandifolia* probably occurred during the Wisconsin full-glacial (18 000 BP) as a scarce tree over much of southeastern North America, and did not have a localized 'refugium'. By 14 000 BP it had begun spreading north, still at very low population densities. Between 14 000 and 10 000 BP it reached high densities locally in the southeast, probably as the result of population expansion while the climate was temporarily favourable. *F. grandifolia* had reached approximately to its present geographic limits by 7000 BP, but continued to spread, at a lower rate, in some areas until the present. The eastern Great Lakes region, where *F. grandifolia* is most abundant today, had the most rapid population expansions, at about 7000 BP, attributed to locally favourable environmental conditions. The spread of *F. grandifolia* across the continent was achieved at very low population densities. The detection and tracking of such a spread is only marginally possible with current pollen-analytical techniques.

Introduction

When the Wisconsin ice-sheets were at their maximum extent roughly 18 000 years ago (BP), the plants which now occur within the glaciated area must have had distributions elsewhere, presumably to the south. Deevey (1949) suggested that the trees of the northern deciduous forests could have had refugia south of the *Picea–Pinus banksiana* forests which prevailed over much of the eastern continent, in peninsular Florida, or in a mixed forest of northern and southern elements along the shore of the Gulf of Mexico. Braun (1950) thought that the vegetation regions south of the glacial border were not displaced during the glacial periods, so that the plants of the glaciated area today have spread north out of a southern forest which has been stable since the Tertiary. Dillon (1956) argued

that the present zones occurred in the same order during the glacial periods, but were compressed towards the south. Palaeoecological studies, however, have shown that 'our present plant communities have no long history in the Quaternary, but are merely temporary aggregations under given conditions of climate, other environmental factors, and historical factors' (West, 1964, p. 55). Recent work has concentrated on elucidating the history of individual species and tracing their changing distributions across eastern North America since the Wisconsin glacial maximum (e.g. Davis, 1976, 1981, 1983), and has established the general directions of movement of forest tree distributions.

The questions that palaeoecologists need to answer about former plant distributions may be listed as follows:

(1) What was the distribution like at the

time of the Wisconsin glacial maximum? (i.e. over what area did the distribution extend, and at what population density?)

(2) What changes in distribution and abundance took place to bring the species distribution to its present state?

(3) At what rate did these changes take place?

Ultimately, we would also like to know how these changes took place, but that can only come by inference from the answers to the descriptive questions above. The better the description of a species distribution change, the more accurate inferences about mechanisms and causes are likely to be. This paper aims to reconstruct the history of *Fagus grandifolia* during the last 18 000 years, using the available pollen data from eastern North America, and to attempt to answer these questions.

The term 'migration', used by many authors to describe the changing distributions of plants during the post-glacial, has been avoided here. To avoid confusion, I suggest that it should be applied only to those movements which have a specific goal, and are usually accomplished in entirety by each individual. Neither condition is true for forest trees, whose distribution changes represent a quite distinct phenomenon from the annual movements of many species of birds, mammals and insects. Instead, the terms 'diffusion' (Pielou, 1979) or 'spread' can be used.

Vascular plant nomenclature follows Gleason (1952).

Mapping past distributions from pollen data

Most of the data on changes in species distribution come from pollen diagrams, which record changes in abundances of several taxa (mostly forest trees, in eastern North America) at one locality over a period of time. Radiocarbon dating provides a useful timescale. Collating data, in various ways, from available pollen diagrams in the area of interest can yield a description of changing distributions and abundances in space and time (e.g. Ritchie, 1976; Bernabo & Webb, 1977; Davis, 1981; Huntley & Birks, 1983; Webb, Cushing & Wright, 1983; Webb, Richard & Mott, 1983).

How can we determine the time of arrival of a taxon near the site of a pollen diagram? We need to distinguish between locally produced pollen and far-travelled pollen, so the answer depends on the pollen productivity and dispersal of the taxon involved, and on what we consider as 'local'. Davis (1981) takes the time at which the curve for a pollen type makes a sharp increase as the time of arrival, inferring that pollen before the sharp rise comes from non-local individuals, and losing all information on abundance. Webb, Cushing & Wright (1983) and Webb, Richard & Mott (1983) have refined the approach of Davis (1981) by determining a pollen frequency for each taxon which they take to indicate, approximately, presence near the site. This is more objective, avoids the problem of determining where the 'sharp increase' is in the rising pollen curve, and is successful in displaying changes in vegetation and ecotones through time. However, like Davis (1981), their approach can still only cope with populations that exceed certain levels. In order to determine how a taxon spread across the continent, information is needed about its status over the entire extent of its range. Identifying areas where it exceeds a certain abundance at each time horizon is useful, but not the whole story.

Many pollen curves increase at an even rate, not showing the 'sharp increase' required by the approach of Davis (1981). Very few studies have looked at the shape of the increasing pollen curves, as Watts (1973a) suggested, but those that have (e.g. Tsukada & Sugita, 1982; Bennett, 1983) have found an exponential or logistic increase from very low values of pollen accumulation rates. Thus, the pollen curve should not be split into two phases as Smith & Pilcher (1973) suggest, and Davis (1981) implies. Davis (1981) also assumes that the 'sharp increase' represents arrival of the taxon. This association is not necessarily accurate, at least in unglaciated areas, where the increase may be simply expansion of populations from low, glacial maximum, levels to higher, interglacial, levels (see the account of *Picea* expansion in northwest Canada by Ritchie, 1984).

Work relating the abundances of trees in forests today to pollen frequencies of the same taxa in surface samples of lake muds

shows a very close geographic relationship (Webb, 1974b; Davis & Webb, 1975; Webb & McAndrews, 1976; Delcourt, Delcourt & Webb, 1984). Very little pollen is dispersed beyond the species range limits for most taxa, at least in forested regions, such as eastern North America. The chances of detecting pollen from trees more than a short distance (2 km) from a small (5–20 ha) lake with no inflowing streams is slight. Such a lake would have approximately 10^6 trees within 2 km (c. 1000 trees ha^{-1}). One tree in this area comprises 0.0001% of the forest. Routine pollen counts rarely exceed 1000 grains per level, so if the taxon has about average pollen productivity and dispersal, it will not be consistently recorded unless its population reached densities of greater than 1 ha^{-1} , or 1000 trees within 2 km of the lake. If the pollen received by the lake comes from a larger area, greater populations of the taxon must occur to reach a level detectable by pollen analysis. For *Tilia cordata* in northern England, 'A large single tree with emergent crown does not contribute significantly to pollen deposited on the forest floor beyond a distance of about 60–100 m' (Pigott & Huntley, 1980, p. 147). A population density of $1\text{--}4 \text{ ha}^{-1}$ would therefore be necessary to achieve detectable pollen of this species everywhere. Thus, the presence of a species around a pollen site, but undetected because of a small population size, is as much a potential source of error as assuming local presence from far-travelled pollen. If a species presence at a particular place and time is known from macrofossil data, but no pollen of the taxon is recovered from contemporaneous sediments, it is not necessary to assume that it is producing little or no pollen, or that it is reproducing asexually (cf. Hopkins, Smith & Matthews, 1981). It may just have had very low population densities (1 tree ha^{-1} or less).

In order to understand the changing plant distributions and abundances of the post-glacial, a flexible approach to the early stages of increase of pollen curves is needed. In some instances they may represent far-travelled pollen, but in others they may represent pollen from small local populations. On the whole, it seems safer to interpret low, continuous, pollen records (at least for the

trees in a forested area) as indicating local presence, unless there seems to be special reason to be wary of far-travelled pollen (e.g. for pollen of *Pinus*). The pollen curve can then be treated as a continuous record of population expansion. The chance of error from far-travelled pollen can be reduced by knowledge of the whereabouts of dense populations of the taxon which could be a source for far-travelled pollen grains. Additionally, macrofossils are usually transported only short distances relative to pollen (Faegri & Iversen, 1975), and may indicate presence of a taxon at a time of very low population density. It is also necessary to recognize that some changes in pollen records may be due to changing abundances of plant populations, and not the result of the extension of species range limits to the area around the site.

Ecology of *Fagus grandifolia*

F. grandifolia is a large forest tree, the only member of its genus native to North America. It ranges across most of the eastern continent (Fig. 1), with outlying populations in the mountains of Mexico (Little, 1965, 1971). The populations in Mexico may have been separated from those in eastern North America throughout the Pleistocene (Martin & Harrell, 1957).

In extreme southern Ontario, *F. grandifolia* is co-dominant with *Acer saccharum* on dry-mesic, mesic and wet-mesic soils (Maycock, 1963). Further north, it occurs almost exclusively on dry-mesic soils with *A. saccharum* and *Pinus strobus* (Lambert & Maycock, 1968), and reaches its northern limit at the southern edge of the boreal forest (Rowe, 1972). Further east, also, *F. grandifolia* tends to occur on drier soils towards its northern limit (Beschel, Webber & Tippet, 1962). In eastern Wisconsin, *F. grandifolia* is co-dominant on mesic soils only, with *A. saccharum* and *Tsuga canadensis* (Curtis, 1959). Further south, Whittaker (1956) found that *F. grandifolia* reaches its highest frequencies in high elevation deciduous forests of the Appalachians on sheltered mesic sites. It tends to be replaced by *Quercus borealis* on less sheltered sites

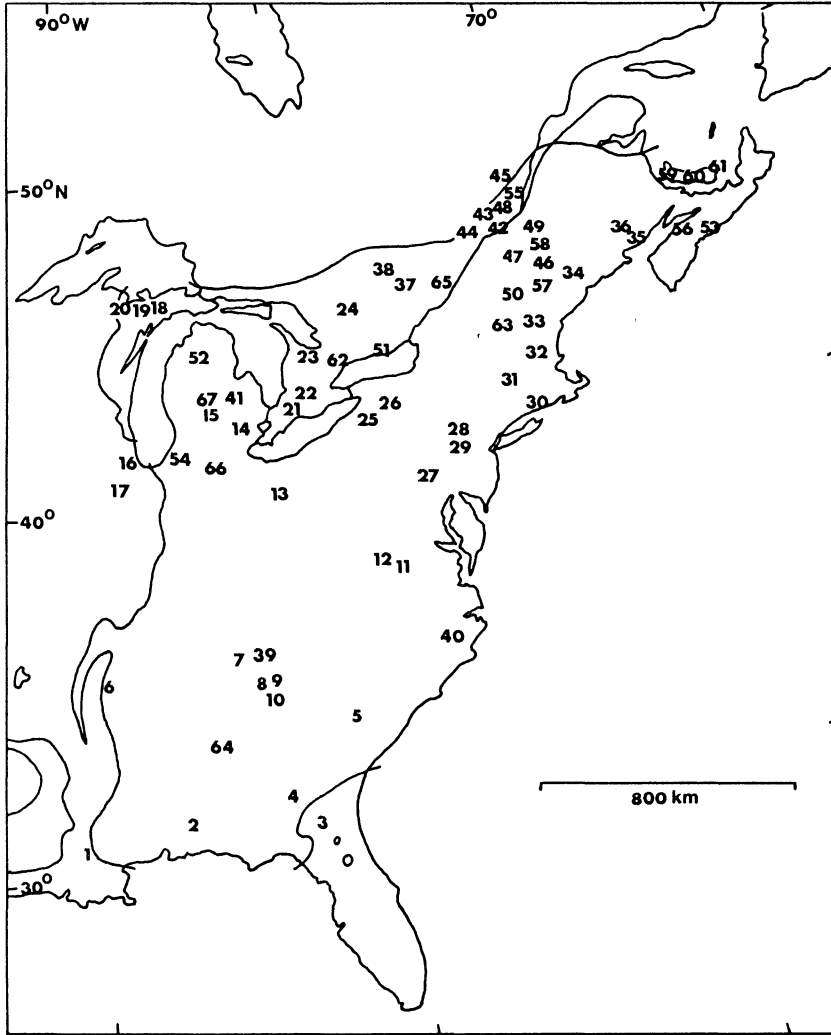


FIG. 1. Outline map of eastern North America showing the location of sites used to construct Figs. 2–6. The continuous line encloses the present distribution of *Fagus grandifolia* in eastern North America (Little, 1971). Numbers refer to authors: 1, Delcourt & Delcourt (1977); 2, Delcourt (1980); 3, Watts & Stuiver (1980); 4, Watts (1971); 5, Watts (1980a); 6, Delcourt *et al.* (1980); 7, Delcourt (1979); 8, Watts (1970); 9, Watts (1973b); 10, Watts (1975a); 11, Craig (1970); 12, Watts (1979); 13, Ogden (1966); 14, Kerfoot (1974); 15, Gilliam *et al.* (1967); 16, 17, King (1981); 18, 19, 20, Davis (1978); 21, McAndrews (1981); 22, Mott & Farley-Gill (1978); 23, 24, McAndrews (1981); 25, Miller (1973); 26, Spear & Miller (1976); 27, 28, 29, Watts (1979); 30, Davis (1969); 31, Whitehead (1979); 32, 33, Davis (1978); 34, Davis *et al.* (1975); 35, 36, Mott (1975); 37, 38, Mott & Farley-Gill (1981); 39, Delcourt & Delcourt (1979); 40, Whitehead (1981); 41, Bailey & Ahearn (1981); 42, 43, Richard (1971); 44, Richard (1973a); 45, Richard (1973b); 46, 47, Richard (1973c); 48, Richard (1975); 49, Richard & Poulin (1976); 50, Richard (1978); 51, Terasmae (1980); 52, R. Lawrenz in Webb, Cushing & Wright (1983); 53, Livingstone (1968); 54, Manny *et al.* (1978); 55, Labelle & Richard (1981); 56, Hadden (1975); 57, 58, Mott (1977); 59, 60, 61, Anderson (1980); 62, McAndrews (1973); 63, McDowell *et al.* (1971); 64, Delcourt *et al.* (1983); 65, Savoie & Richard (1979); 66, Williams (1974); 67, McMurray *et al.* (1978).

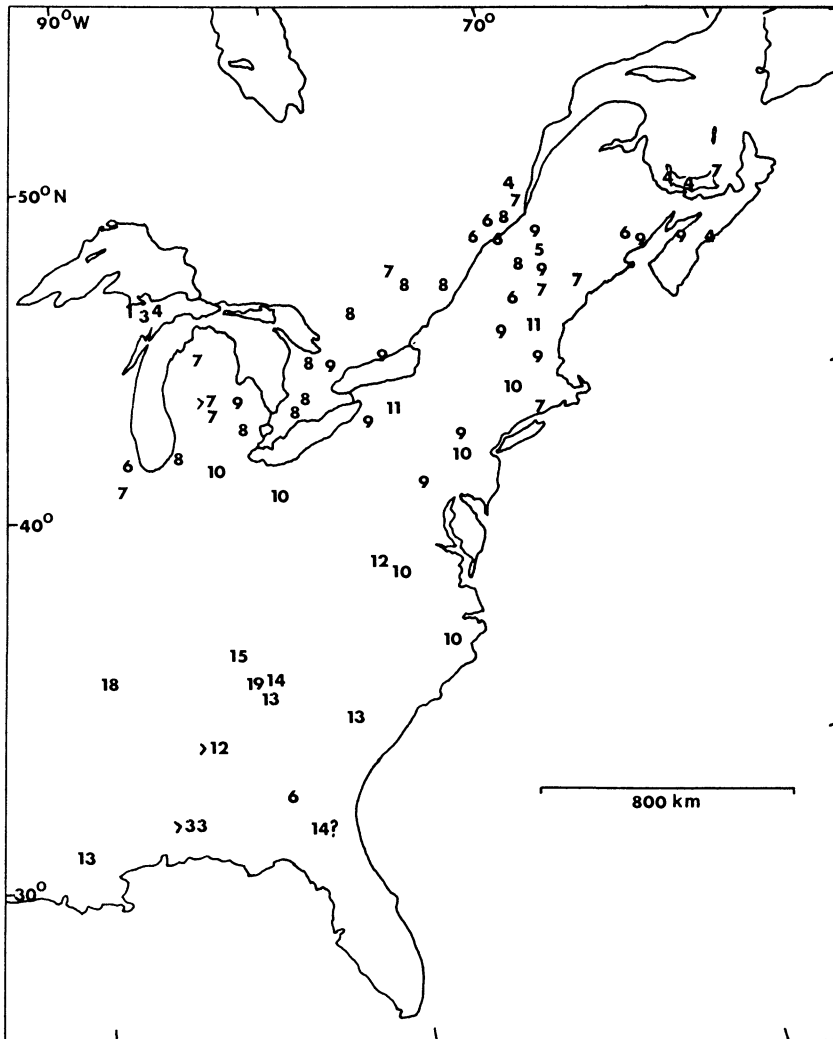


FIG. 2. Dates for the beginning of a continuous curve for *Fagus grandifolia*, in thousands of years BP. Authorities for sites as in Fig. 1.

and moister sites, by *Tilia americana* and *T. canadensis* at lower elevations, and by *Q. alba* and *Castanea dentata* on drier sites. It is also one of the dominants of southern mixed forests from Mississippi to northwest Florida with *Liquidambar styraciflua*, *Magnolia grandiflora*, *Pinus taeda*, *Q. alba* and *Q. laurifolia* (Küchler, 1964).

F. grandifolia is considered to be very shade tolerant (Baker, 1950; Curtis, 1959). It reproduces both by seed and from basal sprouts (Ward, 1958). Patterns of seedling distribution suggest that *F. grandifolia* and

A. saccharum tend to replace each other, while *T. canadensis* and *T. americana* in the same forests are self-replacing (Woods, 1979, 1984). However, McIntosh (1972) found that abundant basal sprouts mean that *F. grandifolia* can also replace itself. Other components of the canopy tend to reproduce only in openings (Curtis, 1959).

Three varieties have been recognized within *F. grandifolia*, differing morphologically and ecologically (Gleason, 1952; Sargent, 1965; Little, 1965). The type variety (var. *grandifolia*) is a tree of rich upland soils in southern

Canada, the Great Lakes region and New England, extending southwards to roughly 37° N, but further in the mountains of North Carolina. Var. *caroliniana*, on the other hand, is a tree of wet lowland soils, such as bottomlands and swamp margins. It extends on the coastal plain from Massachusetts south to northern Florida, west to eastern Texas, and also north in the Mississippi valley to southern Ohio, Illinois and Missouri. Var. *caroliniana* is therefore the tree of the southern mixed forests, whereas var. *grandifolia* is the tree of the northern upland deciduous forests. Var. *mexicana* is the taxon in the Mexican mountains. Additionally, three races have been reported (Camp, 1950), one of which (white beech) appears to be identical with var. *caroliniana* (Little, 1965).

Data

A series of maps is presented displaying the available data in various ways and at three time periods to extract the maximum amount of information, from pollen and macrofossils, on the spread of *F. grandifolia*. The locations and authors of the sites used are given (Fig. 1). Only pollen diagrams with radiocarbon dates relevant to the post-glacial distribution of *F. grandifolia* have been included. No pollen diagram from outside the area of present day *F. grandifolia* distribution (Little, 1971) shows a continuous curve for pollen of *F. grandifolia*. Only one pollen diagram from within that area, on Manitoulin Island, Lake Huron (Warner, Hebda & Hann, 1984), does not have a continuous curve, although *F. grandifolia* does occur in the forests of Manitoulin Island (Morton, 1977). A map of the frequency of *F. grandifolia* in eastern North American forests today, and a map of modern *F. grandifolia* pollen frequencies have been produced by Delcourt *et al.* (1984). These maps represent the situation since the extensive clearance of European settlement, and so cannot easily be compared to the fossil data. The date of the beginning of a continuous curve for pollen of *F. grandifolia* for each site is given (Fig. 2). This time obviously depends to some extent on the number of grains counted by the analyst at

each level, and the number of levels counted, but in general it represents the level at which pollen of *F. grandifolia* begins to consistently exceed 0.1–0.3%. Data from pollen diagrams and macrofossil finds are summarized for the full-glacial (18 000–14 000 BP, Fig. 3), the late-glacial (14 000–10 000 BP, Fig. 4), and the post-glacial (10 000–0 BP, Fig. 5). Pollen of *F. grandifolia* is represented in lake muds roughly equal in proportion to its occurrence in forests around the site (Davis & Goodlett, 1960; Davis, 1963; Livingstone, 1968; Delcourt *et al.*, 1984). Because only sporadic amounts occur beyond the present northern limit (Davis & Webb, 1975; Webb & McAndrews, 1976) and at the western limit pollen percentages and accumulation rates are close to the minimum detectable by routine pollen analysis (Davis, 1978), it is reasonable to assume that low frequencies of *F. grandifolia* pollen do indicate local presence of the tree.

Where maximum frequencies of *F. grandifolia* pollen exceed 5%, the frequency reached is plotted on Figs. 3–5. The date at which *F. grandifolia* pollen frequencies reach maximum levels is also given (Fig. 6). This date might well be slightly different if pollen accumulation rate (PAR) data were used, but percentage pollen diagrams have been preferred here for consistency of treatment (cf. Davis, 1981). Very few PAR diagrams are available, and it is not yet possible to compare PAR for individual taxa from one site to another because of differences in modes of sedimentation (Davis, Brubaker & Webb, 1973; Pennington, 1973; Birks & Birks, 1980).

Discussion

18 000–14 000 BP (Fig. 3)

During this period there are continuous curves for *F. grandifolia* pollen from several sites, all in the western half of the south-eastern United States. Pollen frequencies are all much less than 5%. Macrofossils have been recorded from two sites, one full-glacial in Tennessee (Delcourt *et al.*, 1980), and the other late-glacial from southern Louisiana (Delcourt & Delcourt, 1977). If the dating and stratigraphy is secure (see Watts, 1983),

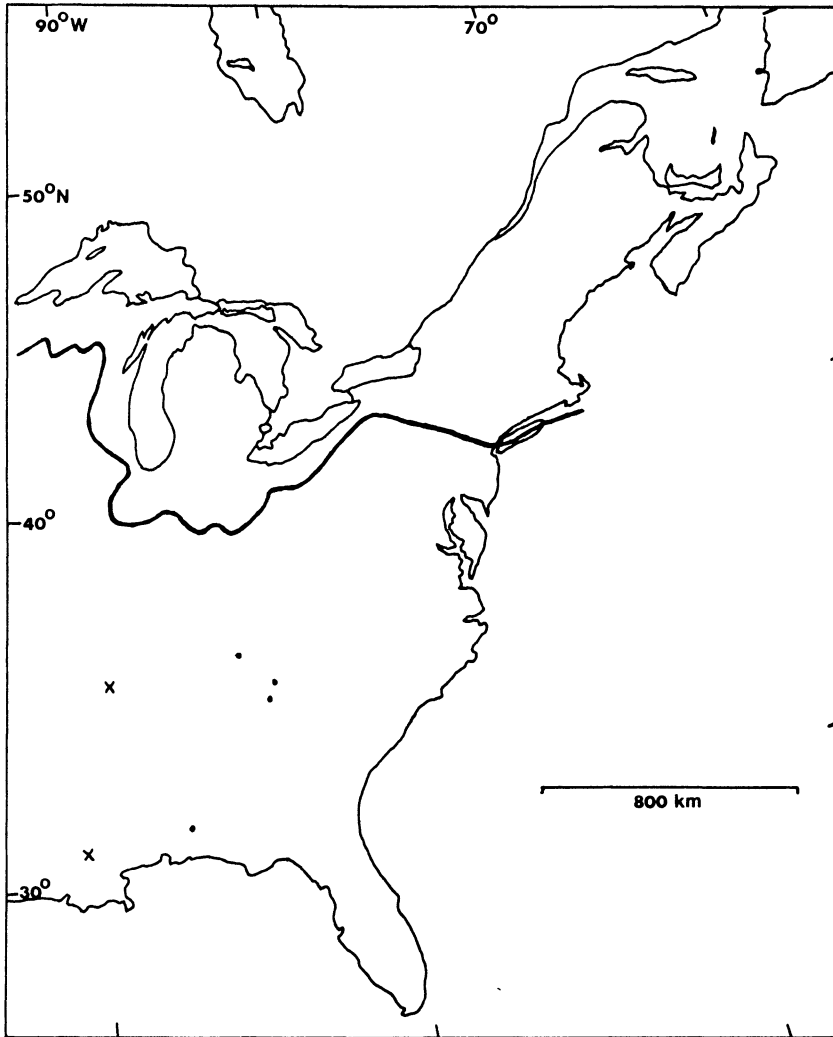


FIG. 3. *Fagus grandifolia* 18 000–14 000 BP. X indicates presence of macrofossils; • a radiocarbon dated pollen diagram with a continuous curve for *F. grandifolia* pollen. The approximate southern glacial limit is indicated by the thick line (Mickelson *et al.*, 1983). Authorities for sites as in Fig. 1.

these macrofossils demonstrate the presence of *F. grandifolia* in the south at the time, while pollen frequencies suggest that, although present over a wide area, it was generally very scarce. Pollen of *F. grandifolia* is actually absent from the horizons associated with the Tennessee macrofossils, although it occurs both above and below (Delcourt *et al.*, 1980).

H. R. Delcourt & P. Delcourt (1975) and P. A. Delcourt & H. R. Delcourt (1977,

1979) have suggested that *F. grandifolia*, along with other deciduous trees of generally northern distributions today, had a full-glacial refugium in the Tunica Hills and the bluffs either side of the Mississippi valley from the Gulf coast at least as far north as Tennessee. This is certainly possible, but there seems to be no reason to suppose that the distribution of *F. grandifolia* was so limited, and the pollen evidence indicates a more widespread distribution (Fig. 3).

The Mississippi valley macrofossils (Delcourt & Delcourt, 1977; Delcourt *et al.*, 1980) are from within the present geographical range of *F. grandifolia* var. *caroliniana*, and also in the same habitat. The populations today of the southern coastal plain and Mississippi valley may be descendants of trees which have occurred there at least since the full-glacial, and the var. *grandifolia* populations of northern uplands may be descended from trees of southern upland areas during the full-glacial. These two varieties have distinctive leaf and cupule morphologies (Gleason, 1952; Sargent, 1965), which ought to be distinguishable in fossil specimens, and it may be possible to trace their history when more fossil localities have been studied.

F. grandifolia may have occurred at very low densities within the prevailing *Picea*–*Pinus banksiana* forest (Wright, 1981; Watts, 1983) quite far north, as it does today in the predominantly coniferous forests of the southern part of the Canadian Shield. Wayne (1967) suggests, from molluscan evidence, that mixed conifer–hardwood forests extended as far north as southern Indiana, about 150 km from the ice margin, during the full-glacial. *Quercus* was a prominent component of the full-glacial coniferous forests at least as far north as southern Illinois (Grüger, 1972a, b). If temperatures there were adequate for *Quercus*, there seems to be no reason to suppose that *F. grandifolia* could not have survived at a similar latitude, as both have similar northern limits today in southern Canada (Little, 1971). Both limits appear to be determined by the minimum temperature to which they can maintain supercooling of intercellular water (-41°C , *F. grandifolia*, *Quercus rubra*, *Q. alba*; -43°C , *Q. macrocarpa*: George *et al.*, 1974; Burke *et al.*, 1976), so it seems reasonable to suppose that they would have had similar northern limits in response to the latitudinal temperature gradient of the full-glacial. The -40°C absolute minimum isotherm in southern Canada today approximates to the northern limit of many deciduous trees (Burke *et al.*, 1976). If this isotherm could be located in North America for the full-glacial, from independent evidence, it could be usefully compared to the full-glacial limits of many

of these trees. *F. grandifolia* is not the only tree which may have been widespread but scarce at this time.

Huntley & Birks (1983) suggest that full-glacial refugia of European trees and shrubs may be identified by locating the region(s) where each taxon first expanded after 13 000 BP. The earliest expansions of *F. grandifolia* are scattered across the southeast (see next section), and so, by the Huntley & Birks (1983) criterion, *F. grandifolia* should have been located in that area during the full-glacial, supporting the argument above.

14 000–10 000 BP (Fig. 4)

F. grandifolia probably began to expand its populations and its distribution as soon as the climate began to warm after the glacial maximum, perhaps soon after 18 000 BP, but we have no evidence for change until soon after 14 000 BP. The pollen record suggests a slightly more widespread distribution for this period than during the glacial maximum. In the southeast, at a number of sites, *F. grandifolia* pollen percentages reach up to 40% in pollen spectra dominated by pollen of *Quercus* and *Carya*. Unfortunately, no site is dated closely enough to be certain of the timing and rates of expansion. Watts (1980b) suggests that this was a transitional forest of a warmer, moister period than the full-glacial, and lasted until the post-glacial climate became too warm and dry for species such as *F. grandifolia*, while *Pinus* spp. increased. *F. grandifolia* pollen reached 7% at Sheelar Lake in north-central Florida (Watts & Stuiver, 1980), at the southern limit of its present distribution. As *F. grandifolia* probably never occurred further south in peninsular Florida (Watts, 1969, 1975b), its distribution in this area seems to have been stable for at least 10 000 years.

We cannot tell whether these increases of *F. grandifolia* in the southeast were the result of range extensions by *F. grandifolia* (presumably from the west) or simply the result of populations expanding as the warming of climate from full-glacial conditions increased the range of sites available. If, as argued above, *F. grandifolia* was wide-

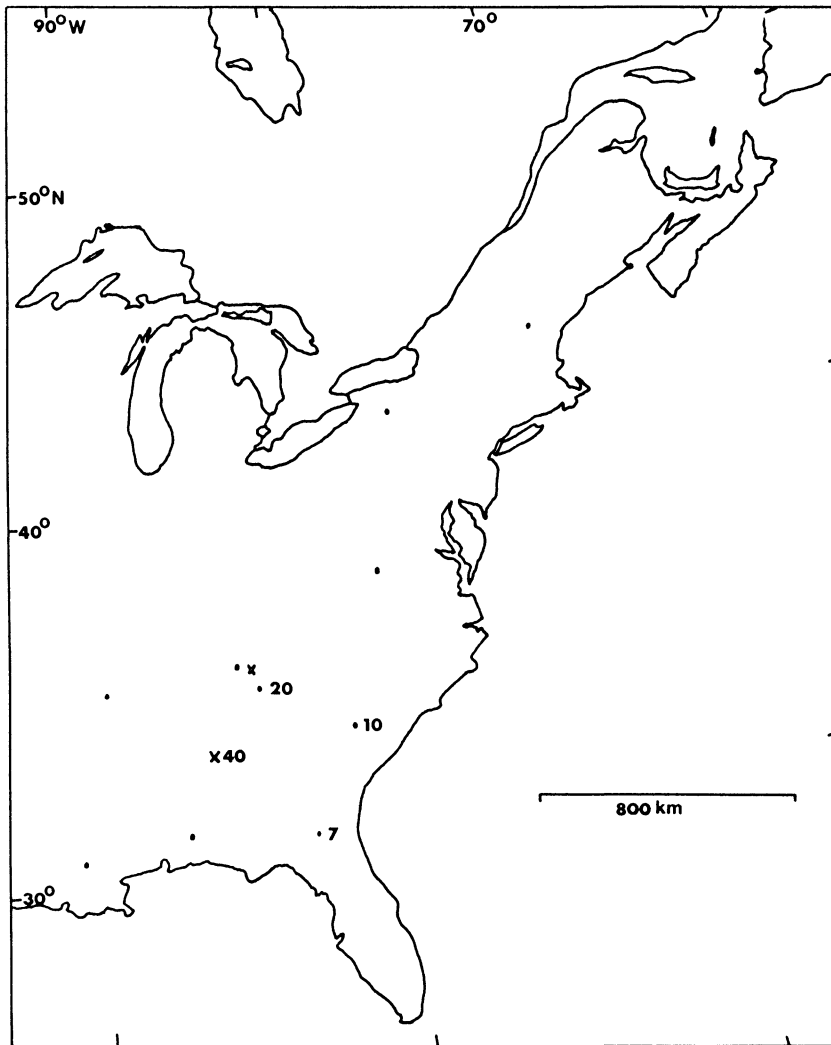


FIG. 4. *Fagus grandifolia* 14 000–10 000 BP. X indicates presence of macrofossils; • a radiocarbon dated pollen diagram with a continuous curve for *F. grandifolia* pollen. Where *F. grandifolia* pollen frequencies exceed 5%, the maximum frequency (as a percentage of total pollen) is given. Authorities for sites as in Fig. 1.

spread but rare during the full-glacial, then these increases have to be considered as the result of population expansions from full-glacial population densities which were too low to be represented in the pollen record.

10 000–0 BP (Fig. 5)

During this period *F. grandifolia* reached its greatest distribution and abundance since the full-glacial. It became an important forest

dominant, reaching 25% of total pollen at several sites. The timing of this spread is not known in detail because, after 10 000 BP and until about 7000 BP, *F. grandifolia* was scarce everywhere and the pollen records show only low values (much less than 5%) which are difficult to interpret.

The distribution of dates for the beginning of a continuous curve for *F. grandifolia* (Fig. 2) is patchy. In some areas, dates differ by several thousand years at adjacent sites.

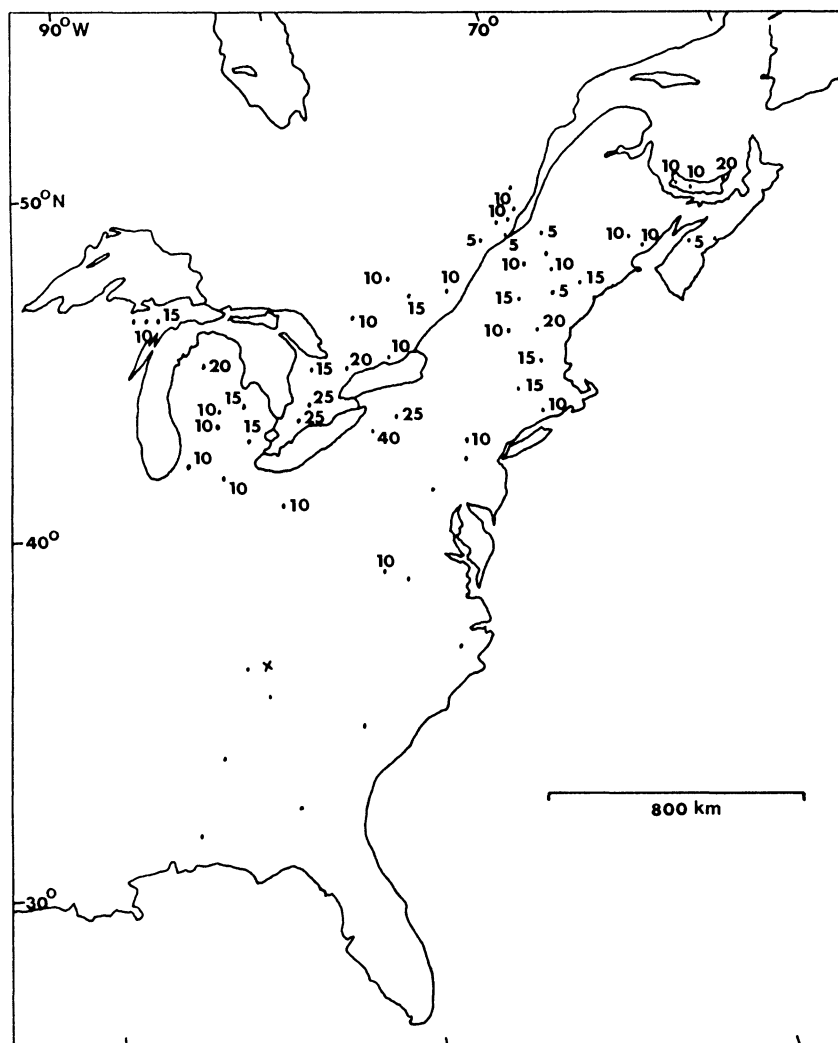


FIG. 5. *Fagus grandifolia* 10 000–0 BP X indicates a macrofossil locality; • a radiocarbon dated pollen diagram with a continuous curve for *F. grandifolia*. Where *F. grandifolia* pollen frequencies exceed 5%, the maximum frequency (as a percentage of total pollen) is given. Authorities for sites as in Fig. 1.

This distribution can be interpreted in three possible ways.

(1) All dates reflect local arrival of *F. grandifolia*. In this case, the spread of *F. grandifolia* was patchy, on the continental scale. Occasional instances of long distance seed dispersal established populations remote from each other, although the intervening areas were suitable for *F. grandifolia*. These populations expanded and eventually coalesced to give the present continuous distribution. Such a model implies an

erratic, but ultimately effective, pattern of seed dispersal, followed by slow population increase.

(2) The older dates in each region are due to far-travelled pollen, or in error, and should be disregarded. In this case, any map of the spread of *F. grandifolia* would have to be based on the later dates, and would give an impression of a slow, steady, spread across the continent, reaching, for example, northern New England at 5000 BP, and the Maritimes at 4000 BP. This model approximates to that

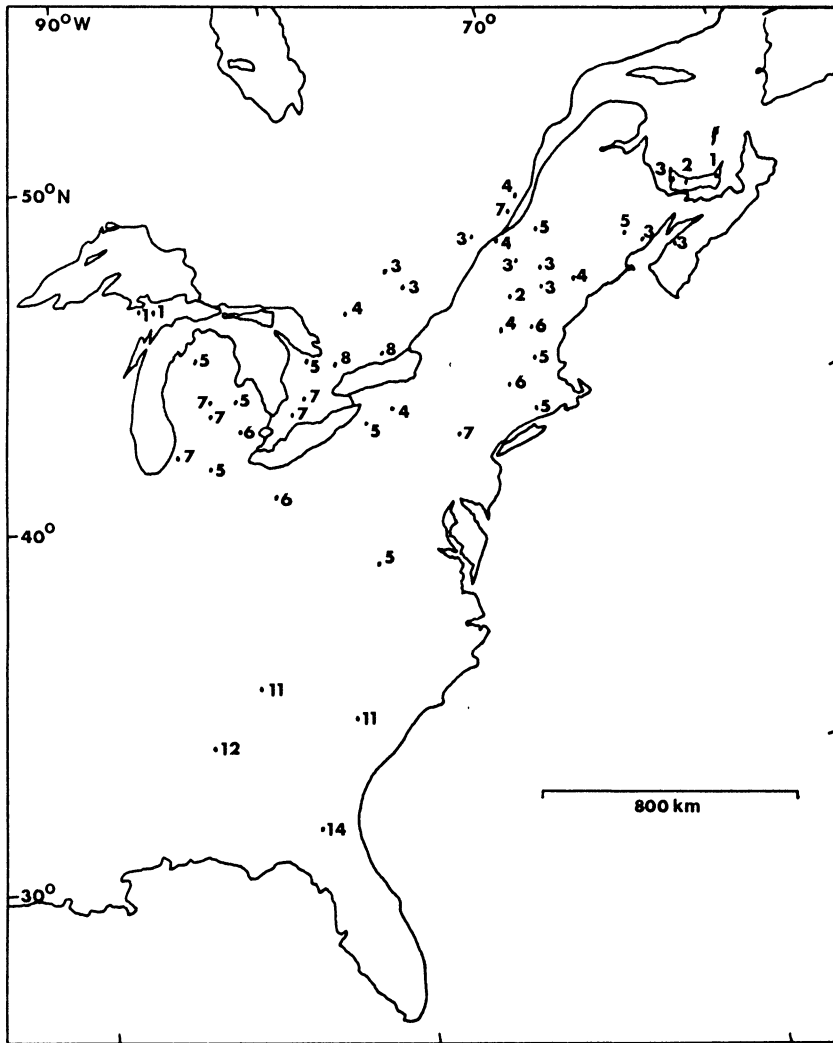


FIG. 6. Date at which *Fagus grandifolia* pollen reaches maximum frequencies, in thousands of years BP. Authorities for sites as in Fig. 1.

adopted by Davis (1981) and implies that seed dispersal was the limiting factor in the spread of *F. grandifolia*.

(3) The younger dates in each region are anomalous, perhaps in areas particularly unfavourable for *F. grandifolia*, and should be disregarded. In this case, any map of the spread of *F. grandifolia* would have to be based on the earlier dates, and would give an impression of a rapid spread at low population densities, reaching, for example, northern New England at 11 000 BP, and the Maritimes at 9000 BP. This model implies that rates

of population expansion after dispersal may be the limiting factors for *F. grandifolia* rather than seed dispersal itself.

No pollen diagram anywhere in eastern North America shows *F. grandifolia* pollen frequencies greater than 5% for the period 10 000–7000 BP, and most have frequencies of much less than 5%. This means that when *F. grandifolia* spread into the Great Lakes region and New England, it must have done so at low population densities. Since the interpretation of pollen data is weakest for low frequencies, it may be that we cannot

properly track the distribution changes during this period. The population densities may easily have been much less than 0.1% in some areas and consequently undetectable. Given the generally low population densities, and hence lack of a source of far-travelled pollen, possibility (2), above, can probably be ruled out. Seed dispersal was probably not the limiting factor in the spread and expansion of *F. grandifolia*. At the moment it is not possible to decide between the others. The possible low densities and slow rates of expansion in some areas make it difficult to decide whether the low frequencies at the beginning of a continuous curve represent arrival within the pollen catchment area of the site or simply the earliest stage detectable in a steady expansion from the even lower frequencies of initial establishment. Hence, we cannot distinguish a patchy spread from a rapid, low density spread: indeed these may be two extremes of a continuum.

The highest frequencies of *F. grandifolia* pollen were reached in the eastern Great Lakes area, and east into New England (Fig. 5). This is the region where *F. grandifolia* is most abundant today (Küchler, 1964; Rowe, 1972). The date at which *F. grandifolia* reached its maximum frequency is variable (Fig. 6), but is generally earliest in the area around Lake Erie and Lake Ontario, where it was most abundant. From this region, to the north, south, east and west, the time between earliest continuous occurrence and time of reaching maximum frequencies (compare Figs. 2 and 6) becomes longer. This suggests that it is the suitability of an area for *F. grandifolia* which determines its rate of population expansion. The time of first occurrence of the tree in a region is a function of distance of seed source and dispersal distances, but the time and rate at which it reaches maximum frequencies depend on the local physical and biotic conditions. Where conditions are favourable, the first, low density, populations of *F. grandifolia* expand rapidly to become large populations. Elsewhere, in less favourable regions, expansion is slower, and the ultimate population size is also lower.

F. grandifolia had reached near to its maximum distribution limits by 7000 BP. It may have continued to spread a short

distance further to the north in the Laurentides, Québec (Richard, 1973b), possibly along the north shore of Georgian Bay (a pollen diagram from near Sault Ste Marie shows a late, unfortunately undated, rise of *F. grandifolia*; Saarnisto, 1974), and into the upper peninsula of Michigan, where expansion may still be taking place (Davis, 1978). In most areas, *F. grandifolia* was present by 7000 BP, but continued to increase its population densities until several thousand years later.

Pollen diagrams from lakes in eastern Wisconsin (West, 1961; Webb, 1982) show curves for *F. grandifolia* expanding in the mid post-glacial up to about 15% of total pollen. Webb (1982) argues that *F. grandifolia* populations of eastern Wisconsin can have arrived there only by long-distance dispersal of the seeds by passenger pigeons (*Ectopistes migratorius*) or blue jays (*Cyanocitta cristata*) either across Lake Michigan or across the prairie peninsula at the southern tip of the lake. Davis (1978) suggests that *F. grandifolia* only arrived in upper Michigan very recently, too late for these populations to have produced those further south. However, a pollen diagram from Volo Bog, near the southern tip of Lake Michigan (King, 1981), has very low frequencies of *F. grandifolia* pollen, so *F. grandifolia* may have spread around the southern end of Lake Michigan at low population densities. The mode of dispersal of *F. grandifolia* seeds is discussed further below. Pollen diagrams from further west in Wisconsin (e.g. Webb, 1974a; Davis, 1977) show no pollen for *F. grandifolia*, and today *F. grandifolia* only occurs in the counties bordering Lake Michigan (Ward, 1956; Curtis, 1959).

Some interesting features are discernible in southern pollen diagrams. At Anderson Pond, Tennessee (Delcourt, 1979), Goshen Springs, Alabama (Delcourt, 1980), north-western Georgia (Watts, 1970, 1975a) and White Pond, South Carolina (Watts, 1980a), the *F. grandifolia* curve, which is continuous during the early post-glacial, becomes discontinuous and often ceases in the mid post-glacial, although within the present range of *F. grandifolia*. On the other hand, a curve begins during the post-glacial (after 6700 BP) and continues to the top of the profile at

Lake Louise, southern Georgia (Watts, 1971). The decrease in *F. grandifolia* is associated with a gradual change in the vegetation from *Quercus* domination to *Pinus* domination, attributed by Watts (1980b) to a slowly changing climate. *F. grandifolia* might have decreased either because the climatic change directly affected it, or because it was less tolerant of *Pinus* spp. as competitors than *Quercus* spp. The association of a rise in *F. grandifolia* pollen frequencies with *Pinus* pollen frequencies at Lake Louise is problematical. Populations at the extreme south-eastern limit of *F. grandifolia* in Florida may be in decline today (Ward, 1967).

At Silver Lake, Ohio (Ogden, 1966) and Pretty Lake, Indiana (Williams, 1974), there is a pronounced minimum of *F. grandifolia* pollen frequencies between maxima at 5000–6000 BP and 1000 BP. This is attributed by Ogden (1969, 1977) to a mid post-glacial dry period ('Xerothermic Interval'), when prairie elements reached their greatest extension eastwards (Webb, Cushing & Wright, 1983), and taxa with greater moisture requirements declined. The second expansion of *F. grandifolia* to the maximum at 1000 BP occurs at the same time as a general decline of *F. grandifolia* pollen frequencies to the north and east (see below).

Several pollen diagrams in the north (e.g. Kerfoot, 1974; Miller, 1973; Davis, 1969; Whitehead, 1979; McAndrews, 1973, 1981; Davis *et al.*, 1975; Spear & Miller, 1976; see also Webb, Richard & Mott, 1983) show declines of *F. grandifolia* pollen from about 2000–1000 BP until the present, and historical evidence suggests a decline in eastern Wisconsin between 1835 and 1953 AD (Ward, 1956). Webb (1973) notes that modern surface samples have less *F. grandifolia* pollen than presettlement samples, but attributes the difference to logging and clearance by settlers. This cannot be a complete solution, as the decline began before European settlement. Other taxa change frequency as well, notably a rise in pollen of *Pinus strobus*, while *Picea* was expanding southwards (Webb, Richard & Mott, 1983). McAndrews (1981) suggests the change might be due to decreasing temperatures, but points out that it could also be the result of Indian agriculture, with field

clearance and subsequent overgrowth causing shifts in the relative abundances of the trees. The widespread nature of the event, including sites on the Canadian Shield (e.g. Found Lake; McAndrews, 1981) seems to make a human cause improbable, but cooling of temperatures may be responsible. The expansion of *F. grandifolia* pollen frequencies to a maximum in Ohio and Indiana at about 1000 BP can also be attributed to a cooling temperature trend in an area which was previously too warm and dry for *F. grandifolia*. *F. grandifolia* has therefore responded in different ways to post-glacial climatic changes in different parts of the continent. It decreased gradually in response to an increasingly arid environment in the south-east (see above), while later increasing in one area and decreasing in another in response to a cooling temperature trend.

Conclusions

It is now possible to suggest, for *F. grandifolia*, answers to the questions posed in the Introduction:

(1) During the Wisconsin glacial maximum, *F. grandifolia* probably extended over a large area of southeastern North America, but at low population densities.

(2) After the glacial maximum, *F. grandifolia* began to spread northwards, towards, and onto, the glaciated area. For several thousand years, until 7000 BP, it was not common (except locally in the southeast, 14 000–10 000 BP), and the record of its spread is poor. It probably reached near to its present range limits by 7000 BP. The eastern Great Lakes area proved to be specially favourable to it, and it increased there to become an important forest tree, co-dominant with *Acer saccharum* over large areas.

(3) As noted above, we know little about its rate of spread, but an estimate can be made. It probably began a range extension at 16 000 BP from a northern limit in Tennessee (a conservative estimate), and it reached near to its northern limit on the Canadian Shield by 7000 BP, a distance of 1400 km, giving an average rate of spread of about 150 m yr^{-1} . Davis (1981) estimated 200 m yr^{-1} . Rates of expansion of popula-

tions seem to vary greatly from one region to another, but have not yet been calculated in detail. The rate of spread may have slowed considerably near the northern limits of the species, as suitable sites became less dense, and hence not located and colonized as readily.

No observations have been made for *F. grandifolia* which would explain the mechanism of spread. Fowells (1965, p. 174) states that 'most of the seed simply drops to the ground under the parent trees. Birds and rodents carry some of them short distances from the source, and on steep terrain, a few may roll down slopes, but dispersal is quite restricted'. *F. grandifolia* does not begin seed production until it is 40 years old (Fowells, 1965). The maximum distance of spread per generation must therefore have been at least 6 km, a figure much greater than Fowells (1965) implies is possible. Bird and mammal dispersal of the seeds is probably much more important than he recognized.

Van der Pijl (1969) distinguishes dispersal inside animals (endozoochory) from dispersal on animal exteriors (epizoochory). For *F. grandifolia*, both processes may be involved. Martin, Zim & Nelson (1961) list twenty-eight species of mammals and birds which consume beechnuts, comprising more than 2% of the diet in thirteen species. In addition, the extinct passenger pigeon is known to have eaten beechnuts preferentially (Schorger, 1973). A small number of species, notably the blue jay, certain woodpeckers (Picidae) and some rodents regularly store nuts for future consumption (Darley-Hill & Johnson, 1981; Bent, 1964; Martin *et al.*, 1961). Of these, the blue jays and woodpeckers are most likely to remove nuts to great enough distances to be a significant element in the spread of *F. grandifolia*, although both seem to prefer acorns of *Quercus*. Of the other taxa, which all consume the nuts as they locate them, probably only the passenger pigeon (Webb, 1982) and deer are large enough to carry seeds over long distances before they are destroyed by digestion, making possible occasional, accidental, dispersal of a viable seed.

On the other hand, the cupules enclosing the nuts bear prickles, up to 10 mm long. Each prickle is covered with stiff hairs, so

that the cupules could become attached to mammal fur. The cupules may fall from the tree with the nuts still inside, so dispersal by epizoochory is possible, particularly on the fur of large mammals (cf. van der Pijl, 1969).

The history of *F. grandifolia* in the last 18 000 years is thus curious, from scarcity for 10 000 years to abundance for 8000 years, with a local short-lived phase of abundance 11 000 years ago. Its overall range has greatly increased, but its population densities in the south are probably no greater now than they were during the full-glacial, despite substantial changes of vegetation [from *Picea-Pinus banksiana* to predominantly *Quercus* or *Quercus-Pinus* (not *P. banksiana*) woodlands]. *F. grandifolia* is clearly able to spread easily at very low population densities, yet has the capacity for rapid increase in favourable regions. These are important attributes for the survival of a species in a period (the Quaternary) of rapidly changing environmental conditions. Part of the *F. grandifolia* distribution is now located in an area (the southeast) where it can survive both glacial and interglacial periods, so survival of the species is not dependent upon a complete shift of distribution twice during each glacial cycle, as taxa which now occur solely within the glaciated area (e.g. *Pinus banksiana*) must do.

We can only speculate about mechanisms of range extensions and population expansion. Both require an excess of successful reproduction over that necessary to maintain populations at a constant level. It is the degree of excess which determines the rate of extension or expansion. The factors which control this include climate, soils and other organisms (competing species, predators, pathogens and parasites). Favourable climate may be responsible for the success of *F. grandifolia* in the eastern Great Lakes area, combined with a lack of competitors in that region, but it is also possible that it is able to grow and reproduce more successfully on immature soils than on soils which were continuously forested throughout the Wisconsin glacial period.

Carter & Prince (1981) argue that there is a threshold density of sites necessary for maintenance of a population in an area.

The distribution limit of a taxon is reached where the rate of removal of suitable sites exceeds the number of sites available in a given time period. They suggest that, since the removal rate of perennials is lower than that of annual plants, the number of suitable sites that must be available for spread is also lower. Trees should, therefore, as long-lived perennials, be able to spread at the lowest densities of all, and have low densities at their distribution limits. The observed pattern of spread of *F. grandifolia* supports these considerations.

This discussion differs from previous accounts of tree distribution change in eastern North America in three important respects. First, it is argued that *F. grandifolia* occurred over a very large area at low population densities during the full-glacial. It did not have a 'refugium' in any particular area. Second, it appears that the initial spread across eastern North America was achieved at the same low population levels, followed by increase to higher population densities in regions with favourable climate and soils. It is not necessary to suppose that this species (or any other) spread as a 'front', invading new territory at high population densities. Species with sparse and scattered populations have occupied the glaciated area and are as widely spread as those with much higher population densities. Those that we focus our attention on are simply the very few which, probably by chance, have located and colonized suitable environments to become abundant in the current interglacial. Third, emphasis is placed on recognizing the distinction between species arrival at a site and species expansion. A species may be said to be present in an area when it is represented by one individual, but we cannot hope to detect such low population densities by pollen analyses. When a pollen curve first becomes continuous and rises, this may represent continuing expansion of the taxon from earlier frequencies too low to be detected in the pollen counts.

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